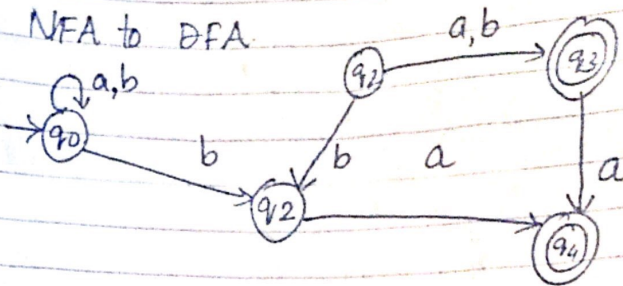


Tutorial-2

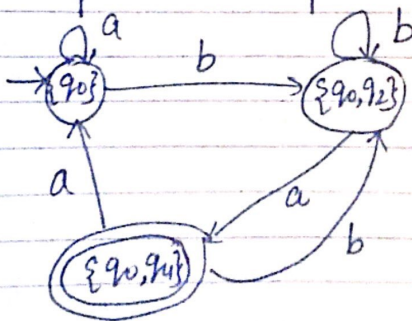
NFA to DFA conversion:-

Q1. NFA to DFA



Transition table of DFA:-

Present state	$\delta/p=a$	$\delta/p=b$
$\rightarrow \{q_0\}$	$\{q_0\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}$	$\{q_0, q_4\}$	$\{q_0, q_2\}$
$\{q_0, q_4\}$	$\{q_0\}$	$\{q_0, q_2\}$



Since q_4 is final state & hence $\{q_0, q_4\}$

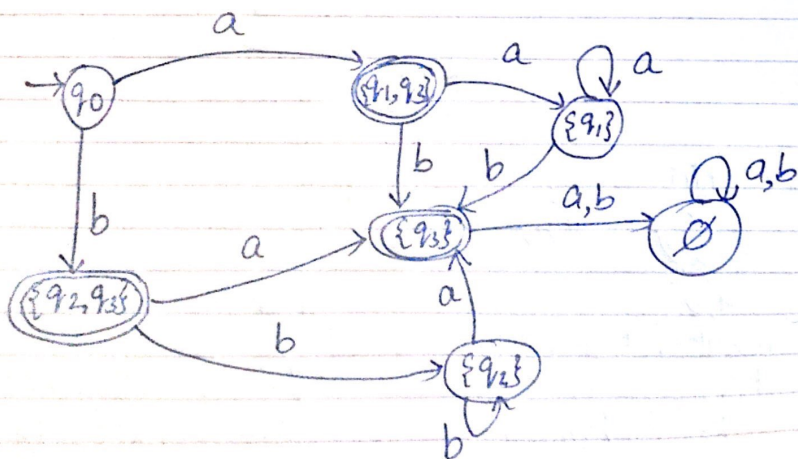
Transition Diagram of DFA

Q2:

State	a	b
$\rightarrow q_0$	q_1, q_3	q_2, q_3
q_1	q_1	q_3
q_2	q_3	q_2
(q_3)	-	-

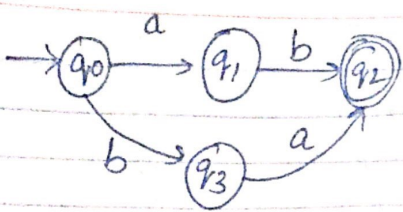
Transition table of DFA

State	a	b
$\rightarrow q_0$	$\{q_1, q_3\}$	$\{q_2, q_3\}$
$\{q_1, q_3\}$	$\{q_1\}$	$\{q_3\}$
$\{q_2, q_3\}$	$\{q_3\}$	$\{q_2\}$
$\{q_1\}$	$\{q_1\}$	$\{q_3\}$
$\{q_2\}$	$\{q_3\}$	$\{q_2\}$
$\{q_3\}$	$\emptyset$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$



DFA

Q3.

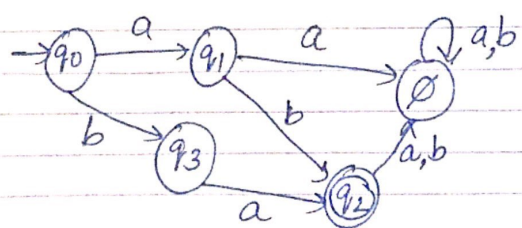


NFA transition table

DFA transition table

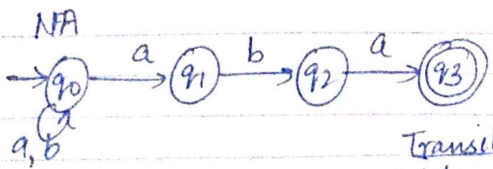
state/ ϵ	a	b
$\rightarrow q_0$	q_1	q_3
q_1	-	q_2
q_2	-	-
q_3	q_2	-

state/ ϵ	a	b
$\rightarrow q_0$	q_1	q_3
q_1	$\emptyset$	q_2
q_3	q_2	$\emptyset$
q_2	$\emptyset$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$



DFA

Q4.

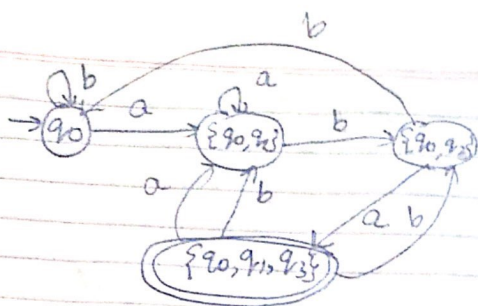


Transition Table of NFA

State/ ϵ	a	b
$\rightarrow q_0$	q_0, q_1	q_0
q_1	-	q_2
q_2	q_3	-
q_3	-	-

Transition table of DFA

state/ ϵ	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	q_0
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0\}$
$\{q_0, q_1, q_3\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$

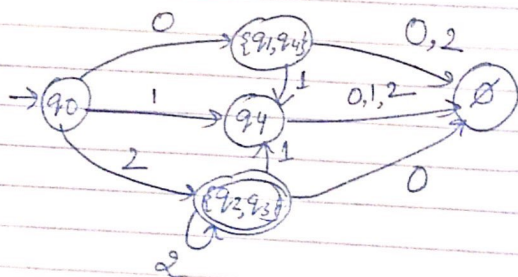


Q5: Transition table of NFA:

State/ ϵ	0	1	2
$\rightarrow q_0$	$\{q_1, q_4\}$	q_4	$\{q_2, q_3\}$
q_1	-	q_4	-
q_2	-	-	q_2, q_3
(q_3)	-	q_4	-
q_4	-	-	-

Transition table of DFA

State/ ϵ	0	1	2
$\rightarrow q_0$	$\{q_1, q_4\}$	q_4	$\{q_2, q_3\}$
$\{q_1, q_4\}$	$\emptyset$	q_4	$\emptyset$
q_4	$\emptyset$	$\emptyset$	$\emptyset$
$\{q_2, q_3\}$	$\emptyset$	q_4	$\{q_2, q_3\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

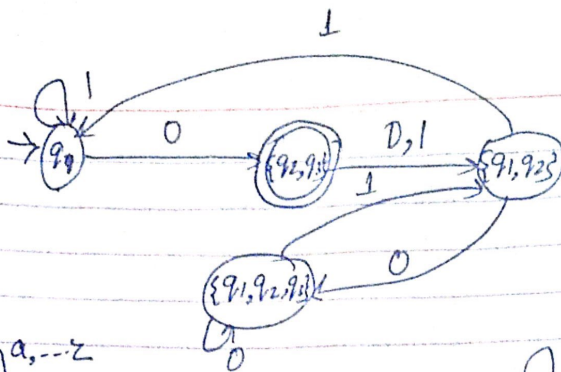


Q6: Transition table of NFA

State/ ϵ	0	1
$\rightarrow q_1$	q_2, q_3	q_1
q_2	q_1, q_2	-
(q_3)	q_2	q_1, q_2

Transition table of DFA

State/ ϵ	0	1
$\rightarrow q_1$	$\{q_2, q_3\}$	q_1
$\{q_2, q_3\}$	$\{q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_1, q_2, q_3\}$	$\{q_1\}$
$\{q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$	$\{q_1, q_2\}$



07.

